Assignment Report

Objective

The objective of this assignment was to build an end-to-end e-commerce order analytics system using Python and SQLite. The project involved generating datasets, cleaning inconsistent data, performing SQL analysis, integrating Python with SQLite, and validating edge cases.

Tasks Performed

1. Data Generation

Generated four datasets:

Customers
Products
Orders
Order Items

The datasets contained intentional inconsistencies such as:

Missing customer IDs
Invalid email addresses
Mixed date formats
Negative quantities
Broken order references

2. Data Cleaning

Implemented the following functions:

clean_orders()
clean_products()
validate_emails()
check_referential_integrity()

The cleaning process standardized date formats, normalized product names, validated email addresses, and identified broken references between orders and order_items.

3. SQL Analysis

Loaded the cleaned datasets into MySQL Workbench and implemented:

Basic SQL Queries

Intermediate SQL Queries
Advanced SQL Queries using CTEs and Window Functions

The queries generated business insights such as revenue analysis, customer rankings, return rates, running totals, cohort analysis, and year-over-year comparisons.

4. Python + SQLite Integration

Created a command-line report generator that:

Accepts a report type and date range
Connects to the SQLite database
Displays:
Total Orders
Revenue
Unique Customers
Top 3 Products
Previous Period Comparison

5. Edge Case Validation

Implemented test functions to validate:

Invalid order_id
Discount greater than 100%
Zero quantity
Future order dates

During validation, 30 invalid order_id references were detected. The assignment required only identifying broken references through check_referential_integrity(), which was implemented accordingly. Hence, the invalid records were reported but not modified. In a production ETL pipeline, these records would typically be corrected or removed before loading the cleaned data into the database.

Conclusion

This assignment provided hands-on experience with data generation, data cleaning, SQL querying, SQLite database management, Python-SQL integration, and data validation. It demonstrated the complete workflow involved in building a basic data analytics pipeline using Python and SQL.